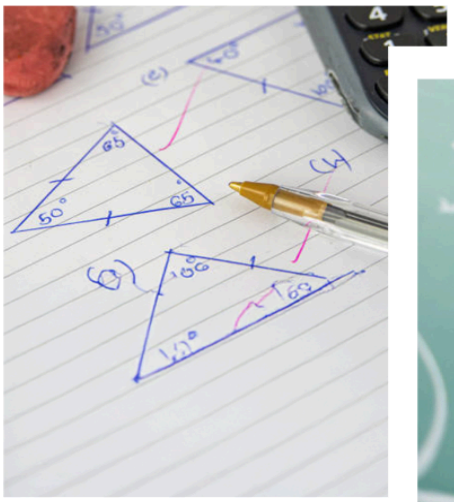
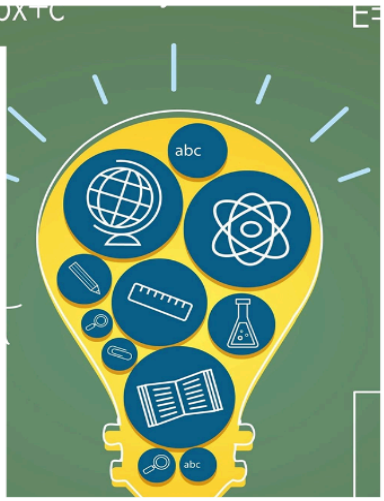
testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Probability





Probability

Outcomes of different sample

Tossing a coin:

- When one coin is tossed then,

  S = { H, T}

- When two coins are tossed then,

  S = { HH, TT, HT, TH }

- When three coins are tossed then,

  S = { HHH, TTT, HTT, THT, TTH, THH, HTH, HHT }

- When four coins are tossed then,

S = { HHHH, TTTT, HTTT, THTT, TTHT, TTTH, HHHT, HHTH, HTHH,

THHH, HTHT, THTH, TTHH, HHTT, THHT, HTTH }

- Total possible outcome of tossing a coin n times = 2^n

Throwing die:

- When a die is thrown once then S = {1, 2, 3, 4, 5, 6},

  n(S) = 6

- When two dice are thrown together or A die is thrown twice then

  S = { (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6) (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6) (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6) (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6) (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6) (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6) }

  n(S) = 6 × 6 = 36

- When 3 dice are thrown or a die is thrown thrice

  Then, n(S) = 6 × 6 × 6 = 216,

  n(S) → no. of outcomes in sample space.

- Total possible outcome of throwing a die n times or n dice are thrown together $= 6^n$

Deck of Cards:

- There are 52 cards in a standard deck.
- There are 4 suits; Diamonds, Clubs, Hearts, Spades.
- "Face Cards" include Jacks, Queens, and Kings. (Total face cards = 12)
- Black cards include all Clubs and Spades. (Total black cards = 26)
- Red cards include all Hearts and Diamonds. (Total red cards = 26)

Thus, there are 52 cards in a deck. Hence, the sample space is all 52 cards.

When a coin and a dice are tossed together, the corresponding sample space for the random experiment:

S = { (H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6) }

Mutually exclusive event

Let A and B are two mutually exclusive events, then

- $P(A \cap B) = 0$
- $P(A \cup B) = P(A) + P(B)$

Independent events

If A and B are independent events then,

$P(A \cap B) = P(A) \times P(B)$

Probability P(A) $= \frac{m}{n} = \frac{Number\ of\ outcomes\ favourable\ to\ A}{Number\ of\ total\ outcomes}$

Note : $0 \le P(A) \le 1$

Odds in favour and Odds in against

- The odds in Favor of the event E = $\frac{P(E)}{P(E')}$
- The odds against the event E = $\frac{P(E')}{P(E)}$
- If odds in Favor of the event E = a : b , then P(E) = $\frac{a}{a+b}$
- If odds against the event E = a : b , then P(E) = $\frac{b}{a+b}$

Boole's Inequality

- For any two events A and B

  (i) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

  (ii) $P(A \cup B) \le P(A) + P(B)$, $\{P(A \cap B) \ge 0$

- In general, for any n events A1, A2 ..... An

  $P(A1 \cup A2 \cup \ldots \cup An) \le P(A1) + P(A2) + \ldots + P(An)$

Conditional probability

- **P(A | B) =** $\frac{P(A \cap B)}{P(B)}$
- **P(B | A) =** $\frac{P(A \cap B)}{P(A)}$

Note : If A and B are independent events, then $P(A \cap B) = P(A)\,P(B)$

Therefore, $P(A \mid B) = P(A)$

Similarly, P(B | A) = P(B)

Bayes Theorem

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{k=1}^{n} P(E_k)P(A|E_k)}$$

- If A and B are two events, then the formula for the Bayes theorem is given by:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad \text{where} \quad P(B) \neq 0$$